FINAL PRESENTATION SCRIPT

Suitability Analysis for Rainwater Harvesting Site Planning Using GIS — Group 11

Measured length: 2,389 spoken words — about 16 minutes at a normal presentation pace (150 words per minute). Delete slide 4 as recommended below and it comes to 2,350 words, a little under 16 minutes. Allow 15–16 minutes and you will not be rushing.

All slide numbers below refer to **the exported PDF deck (32 slides)**, which is what your speaker split uses. Note that `FINAL_PRESENTATION_SPEAKER_NOTES.md` describes 33 slides — it includes a "Progress Since the Last Review" slide that is not in the final deck, so from slide 20 onward the notes run one number ahead. Everything here is already mapped to the deck.

Speaker allocation and time budget

Speaker	Slides	Time
Akshay	1–9	4:00
Yesha	10–17	3:40
Arya	18–24	3:25
Aditya	25–32	4:00

Slide 1 was not in your list. Akshay takes it, since he speaks first.

Per-slide budget

Slide	Sec	Slide	Sec	Slide	Sec	Slide	Sec
1	20	9	35	17	25	25	30
2	40	10	25	18	35	26	45
3	25	11	25	19	25	27	45
4	15	12	15	20	30	28	45
5	25	13	30	21	30	29	20
6	15	14	45	22	25	30	25
7	45	15	25	23	30	31	25
8	20	16	30	24	30	32	5

Five slides must never be cut if you run late: 7, 14, 26, 27, 28. They carry the entire contribution. If you are behind, take the time out of slides 3, 12, 19, 24 and 29 instead.

A note on pace. This is a 33-minute deck compressed into 15 minutes. The script below is written to be spoken at a normal, unhurried rate — the compression is already done, so do not rush it. Say what is written and stop. Resist adding.

AKSHAY — SLIDES 1 TO 9

1 Title

SCRIPT

> Good morning. We're Group 11. Our project is a GIS-based decision-support system for choosing where to build rainwater harvesting structures in Chandrapur district, Maharashtra. I'm Akshay, presenting with Yesha, Arya and Aditya. Our guide is Mr. Chirag Desai.

TRANSITION

"Let me start with the problem — and there are actually two of them."

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2 Problem Definition

SCRIPT

> Chandrapur isn't short of rain — about fifteen hundred millimetres a year. The problem is timing: it falls in one short monsoon and most of it runs off. So siting is what matters. A structure on steep ground, in poor soil, or far from demand gives no recharge and wastes public money, and six judgements about every location across eleven thousand square kilometres can't be made by hand.
>
> The second problem is methodological. Three weaknesses recur in GIS-MCDA: scores get stretched to fit local data until a layer carries no information, weights are asserted rather than tested, and the method's own influence is never isolated. We found all three in our own work.

OPTIONAL — IF TIME PERMITS

> The four tiles: 11,443 km² district, 123,555 grid cells of about 300 metres each, six criteria, and one ranked shortlist of 25 sites.

LIKELY QUESTIONS

Q: Where does the 1,493 mm figure come from? A: CHIRPS satellite rainfall — eleven annual rasters from 2014 to 2024, averaged. Slide 21 covers how we built it.

TRANSITION

"So why does this matter enough to build a system for it?"

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3 Motivation and Necessity

SCRIPT

> Rain genuinely is lost — at a hundred-millimetre storm the district's mean runoff coefficient is 0.354, and built-up land reaches 0.61. Traditional siting considers factors one at a time. And open data is now good enough that the method transfers to any district for free. The last point shaped our design: planners need reasons, not just a map.

TRANSITION

"That principle came out of what we found in the literature."

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4 Literature Review (GIS and AI agents)

SCRIPT

> These four papers come from our early scoping, when the project included a natural-language query layer. We removed that component — it added no analytical value to siting — so they informed our scoping rather than the final system.

⚠ **Read this before presenting**

> This slide describes a component the project no longer has, and slide 20 says it was removed. **The cleanest fix is to delete this slide before the review.** If it stays, the script above is the honest framing — say it in one breath and move on. Do not describe the NLP layer as current.

TRANSITION

"The papers that actually shaped the methodology are these."

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5 Literature Review (RWH studies, part 1)

SCRIPT

> Eight studies are our real foundation. The pattern is consistent: GIS with AHP and weighted overlay is the standard approach, and it works. Seif reports 18.85% suitable, the Moulouya study validates at an AUC up to 0.93. We took the workflow from these. What we questioned is how carefully each step inside it gets justified.

TRANSITION

"The second half of the set adds the elements we built on directly."

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6 Literature Review (RWH studies, part 2)

SCRIPT

> The Wasit study matters most — it's the only one running a sensitivity analysis, at 34.33% suitable. Awi Zone confirms the criteria we use, and the semi-arid Ethiopia study found slope most influential, which we can now test against our own findings.

TRANSITION

"Reading these eight together, seven gaps stood out."

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7 Research Gaps and How This Project Responds

SCRIPT

> Seven gaps and our response to each. Weights asserted, never tested — three sensitivity analyses plus a matrix the user edits live. One method used, influence unknown — four MCDA engines over one byte-identical criteria stack. Scores normalised to the local range so they don't transfer — fixed literature-based breakpoints, so four degrees of slope scores the same anywhere. Validation absent — ours too; the harness is built but pending an inventory, and we claim no number. Static models — five planning scenarios. Hard to reproduce — open data and provenance in every export. Socio-economics missing — settlement proximity is weighted, cost data is absent and we declare that.

TRANSITION

"Those gaps became our objectives."

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8 Objectives and Current Status

SCRIPT

> Seven objectives, six achieved: one common grid, six criterion layers, AHP weights at a consistency ratio of 0.0235, a classified map on fixed thresholds, twenty-five ranked sites each with its justification, and robustness quantified four ways. The seventh, validation against existing structures, is not achieved. The harness is written; the inventory doesn't exist.

TRANSITION

"Here's how the system is put together."

9 Proposed System and Architecture

SCRIPT

> The architecture splits in two, and that's our most important design decision. Stage one is everything depending only on the data — build the grid, mask, reproject, fill rim gaps, derive drainage density and settlement distance, score all six criteria. Nine-tenths of a second, and cached. Stage two depends on the weights: aggregate, gate, classify, rank, in about 0.15 seconds. Speed is the obvious benefit. The real reason is scientific — every algorithm scores the same cached rasters, so any difference comes from the method alone.

LIKELY QUESTIONS

Q: Why no database or API? A: The analysis is a pure function of files on disk with one user at a time. A database would add operational complexity without buying anything.

TRANSITION

"Yesha will now take you through the study area and the data behind all of this."

YESHA — SLIDES 10 TO 17

10 Study Area: Chandrapur District

SCRIPT

> Chandrapur is in eastern Maharashtra, officially 11,443 square kilometres. We analysed 11,412 — 99.7%. That's a check rather than a statistic: we work in latitude and longitude, where cell width changes with latitude, so matching the official area confirms both the boundary mask and our area arithmetic. Two things shape everything downstream — the district is very flat, and rainfall is monsoon-dominated.

LIKELY QUESTIONS

Q: Why work in lat-long rather than a metric projection? A: It avoids a second resampling of every layer. The latitude corrections are explicit in the code, and the 99.7% area match validates them.

TRANSITION

"Seven datasets feed that grid."

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11 Data Sources

SCRIPT

> Seven open datasets. Two things here are correctness decisions. Resampling depends on data type — continuous bilinear, categorical nearest-neighbour, because averaging soil class four and class six gives class five, which may not exist. And gap filling: the layers were clipped separately, so gaps inside the district are filled from the nearest valid cell — 2.2% for soil, zero for rainfall.

LIKELY QUESTIONS

Q: Isn't gap filling inventing data? A: It's nearest-neighbour substitution over a few hundred metres on layers that vary slowly in space, affecting at most 2.2% of the district. The alternative was discarding the entire rim.

TRANSITION

"These are the six layers that result."

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12 Criterion Layers

SCRIPT

> Six layers on one grid, and two of them we compute rather than download. Drainage density: river lines split into hundred-metre segments, each length accumulated into its cell, convolved with a 2.5-kilometre window. Settlement proximity: cells above 400 people per square kilometre form a mask, and a distance transform gives every cell its distance.

TRANSITION

"Each of those six carries a weight."

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13 The Six Criteria

SCRIPT

> Rainfall carries the largest weight, 0.43 — it's the water source. Slope 0.23, because it governs retention and earthwork. Soil 0.15, because recharge needs infiltration and stability together. Two scoring choices worth noting: soil peaks at Silt Loam rather than Sand, since sand infiltrates fastest but retains nothing; and settlement proximity peaks at three kilometres, because you can't build inside a town and a remote site serves nobody.

TRANSITION

"How each criterion gets scored is where our first contribution sits."

14 Criterion Scoring Methodology

SCRIPT

> Four scoring forms: anchored ramps for slope and drainage, categorical lookups for soil and land cover, percentile rank for rainfall, and a moderate-optimal curve for settlement.
>
> Why we changed this is on the right. Min–max stretches each layer between its own extremes. On our data that left 97.4% of slope cells and 97.7% of population cells above 0.90 — both effectively constant, with slope carrying 22% of the weight and contributing nothing. It was worse for categorical layers: min–max scored them by class code, so water ranked best and forest worst. The map was recommending lakes. Fixed reclassification fixes both and makes the scoring transferable.

LIKELY QUESTIONS

Q: Rainfall uses percentile rank, which is district-relative. Doesn't that contradict transferability? A: It does, and it's the one exception. The whole district gets adequate rainfall, so absolute bands would collapse the criterion. We chose within-district discrimination over portability for that one layer, and we say so.

TRANSITION

"Once the criteria are scored, they need weights."

15 AHP Weighting

SCRIPT

> AHP turns expert judgement into numbers through pairwise comparison on Saaty's one-to-nine scale. Our consistency ratio is 0.0235 against a threshold of 0.10. The point on the right we'd rather make ourselves: a low ratio shows the judgements are internally consistent, not that they're correct. That's exactly why we stress-test them three ways later.

TRANSITION

"With scores and weights, we can compute suitability."

16 Suitability Calculation

SCRIPT

> Three steps. Aggregation is a weighted sum, but only over criteria that have data at that cell, with weights renormalised per cell. That creates a loophole, which the coverage gate closes — a cell is scored only if 80% of the weight has real data behind it. Without it, an edge cell scored on two criteria could outrank a site scored on all six, and thirteen of our original top twenty-five were exactly that. Then fixed thresholds, because a percentile split reports about a third of any district as suitable whatever the scores are.

TRANSITION

"The last step turns the surface into a shortlist."

17 Candidate Site Selection and Explainability

SCRIPT

> Three stages. Best cell from each of twenty-five tiles, so the shortlist isn't clustered in one belt. Remaining slots from district-wide maxima, five kilometres apart, since two structures five hundred metres apart harvest the same catchment. Then re-rank strictly by score — ranking by selection order once published rank 22 above rank 13. Every site carries its per-criterion scores, weighted contributions and raw measured values, so you can argue with it.

TRANSITION

"Arya will take you through what the system actually produced."

ARYA — SLIDES 18 TO 24

18 Result: Suitability Zones

SCRIPT

> 18.51% of the district is highly suitable — 2,112 square kilometres. The surface runs from 0.256 to 0.984 across all 123,555 cells, with exactly one cell at the maximum. That sits just below the 18.9 to 34.3% range in our reviewed studies, and we did not retune thresholds to land inside it. Because the thresholds are fixed rather than percentile-based, the figure measures this district rather than our method.

LIKELY QUESTIONS

Q: Why is your suitable area lower than the published studies? A: Continuous scoring scores lower than step classes that award a full 1.00 across an entire band. We chose not to adjust the thresholds to compensate.

TRANSITION

"From that surface we extract the shortlist."

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19 Result: Top-Ranked Candidate Sites

SCRIPT

> Twenty-five sites, 0.984 down to 0.866. Rank one: slope 0.22 degrees, Silt Loam, cropland, six kilometres from a settlement — measured values at that coordinate, not estimates. The honest observation is that all five leading sites are Silt Loam cropland on flat ground. That's the model behaving consistently with its own tables, but it concentrates the shortlist in one landscape type.

LIKELY QUESTIONS

Q: All your top sites have identical soil and land cover. Isn't that suspicious? A: Silt Loam covers most of the district and scores 1.00, and cropland also scores 1.00. It's a fair criticism of compensatory aggregation — TOPSIS, which penalises weakness harder, still retains nine of our top ten.

TRANSITION

"None of those numbers were true six months ago. This is what changed."

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20 What Was Corrected, and Why It Mattered

SCRIPT

> Eight corrections; three changed the answer most. Min–max normalisation left two criteria constant and scored categories by code number. The drainage criterion was measuring distance to the nearest river, a convenience measure — the literature specifies density, because that tells you where runoff concentrates. And population was scored as 'fewer people is better', which pushed sites into empty forest. The impact line matters: sites scored on all six criteria went from twelve of twenty-five to twenty-five of twenty-five. We found all of these ourselves.

LIKELY QUESTIONS

Q: You found bugs in your own system — doesn't that undermine your results? A: The opposite. Finding them is why the current results can be trusted. Every one was found by checking the implementation against the documented methodology.

TRANSITION

"The biggest single data change was rainfall."

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21 Real Rainfall Data: CHIRPS v3.0

SCRIPT

Rainfall carries the largest weight, and our layer was a single test month — so every earlier result was provisional. It's now an eleven-year mean built in code, with source and units written into the file's own metadata, and the build fails loudly if the mean falls outside twenty percent of the published normal. One finding we state rather than hide: our mean sits about twenty percent above the 1,245-millimetre long-term normal. It isn't a unit error — three years sit near the normal and the rest run higher. The decade is genuinely wetter.

OPTIONAL — IF TIME PERMITS

The data trap: CHIRPS ships minus-9999 as its fill value but does not declare it in the file header, so a standard reader treats it as a real measurement. On one global file that produces a raw mean of minus 6,947 millimetres against a true 889.6. Treating anything below minus 100 as NoData is what prevents it.

TRANSITION

"The drainage criterion needed rebuilding too."

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22 Drainage Density and Derived Hydrology

SCRIPT

River lines split into hundred-metre segments, lengths accumulated per cell, convolved with a 2.5-kilometre window. Verified: 2,765.6 kilometres binned against 2,767.7 of input geometry. The window size is a judgement — one kilometre gives speckle, much larger converges to the district mean. Two deliberate decisions: runoff is not an AHP criterion, because it's a function of rainfall, soil and land cover, all already weighted. And it's reported for a design storm, never annually, because the method is an event model.

TRANSITION

"One more correction, and it's the one that made ranking possible at all."

---

23 Continuous Scoring and Meaningful Rankings

SCRIPT

Every criterion originally had a flat top — slope scored a full 1.00 below one degree, rainfall for the whole wettest quintile. Any cell in the best band of all six scored exactly 1.000, and 973 cells did, so the ranking conveyed nothing. The fix was linear interpolation between the same anchors; the anchors didn't move, only the behaviour between them. Cells at the maximum went from 973 to one. The High share fell to 18.51% as a result, and we didn't retune the thresholds.

TRANSITION

"On top of all this sits the decision-support layer."

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24 Decision-Support Layer

SCRIPT

Five scenarios, each a separate matrix over the same six criteria, each passing its own consistency check. No new data, and rescoring takes a fifth of a second. They change the answer materially — maximising recharge takes the High share from 18.5 to 30%. Community-focused lands at almost the same share but selects a different top site, which is why an area statistic alone isn't enough. Two are labelled proxies deliberately: low-earthwork isn't a cost model, and fertile farmland isn't a fertility model.

TRANSITION

"Aditya will now take you through the comparative work, which is where the main findings are."

ADITYA — SLIDES 25 TO 32

25 Four MCDA Algorithms on One Criteria Stack

SCRIPT

> Two independent choices, and most published work varies only one. First, how weights are derived — crisp AHP treats each judgement as an exact number, Fuzzy-AHP as a range. Second, how scores are combined — WLC is a weighted sum and fully compensatory, while TOPSIS measures distance to a theoretical ideal, so weakness on a heavily weighted criterion is punished disproportionately. Crossing them gives four variants on the same data and grid. One decision worth defending: our first Fuzzy-AHP returned weights identical to crisp AHP, because a uniform multiplicative spread cancels during normalisation.

TRANSITION

"Here is what running all four produced."

---

26 Algorithm Comparison Results

SCRIPT

> Read the first column down. The two WLC rows sit near 18.5%; the two TOPSIS rows near 13%. Now read across the weighting pairs — crisp to fuzzy barely moves anything.
>
> That's finding one, and it's our strongest result. Changing the aggregation rule moves the high-suitability area about five percentage points and drops class agreement to 75%. Changing the weighting method moves it 0.2 points at 98.7% agreement. Studies comparing only weighting methods may be comparing the less consequential choice.
>
> Finding two is subtler. Fuzzy-AHP with WLC correlates 1.000 with our baseline, yet shares only seven of its top ten sites. Candidate sites score very close together, so a small weight change reshuffles them without visibly changing the map.

LIKELY QUESTIONS

Q: Which algorithm is best? A: We can't say, and we won't. Ranking them requires ground truth we don't have. What we can say is that they differ measurably, and that fifteen sites are chosen by all four.

TRANSITION

"The next question is whether the weights themselves hold up."

---

27 Sensitivity Analysis: Three Independent Tests

SCRIPT

> The standard criticism of AHP is that we chose the weights. Rather than argue, we measured it three ways. One-way perturbation: each weight moved twenty percent, ninety percent of the top ten retained, maximum rank shift three. Map removal: delete each criterion and re-score. Effective weights: assigned weight against the share actually contributed.
>
> The three agree on something we didn't expect. Drainage carries 9.3% of the weight and barely moves the map, but removing it leaves only two of the top ten standing, because it has the highest spatial variance of the six. Rainfall goes the other way — removing it inflates High to 42%, because percentile scoring fixes its mean near 0.6.

LIKELY QUESTIONS

Q: Why weight rainfall highest if it barely varies? A: It's the water source, so it belongs at the top of the hierarchy. But we agree its realised influence is lower, and we measured exactly how much lower rather than assuming.

TRANSITION

"Which brings us to the honest question about accuracy."

28 Validation Status and Cross-Method Consensus

SCRIPT

> The validation framework is built — ROC-AUC from the Mann-Whitney U statistic, verified against five known cases. It has never run on real data. No register of existing structures exists for Chandrapur, so every entry point returns 'missing data' and the module won't invent a number. We report no AUC and no claim that one algorithm outperforms another.
>
> What we offer instead is consensus. A site chosen by one configuration might be an artefact of it; a site chosen by all four is not. Fifteen of thirty-six locations were selected by all four algorithms, and 1,417 square kilometres was classed highly suitable by every one. I'll say it before anyone asks: consensus is agreement, not correctness.

LIKELY QUESTIONS

Q: How do you know your map is accurate? A: We don't, and we say so. We verified the software — eleven automated checks, a 99.7% area cross-check, a 0.08% river-length conservation check. We did not validate the map against reality, because the data to do that doesn't exist.

TRANSITION

"Briefly, the application itself."

29 The Application and How It Was Verified

SCRIPT

> A Streamlit application with a Folium map — data-status checks, the scenario picker, the editable matrix, site cards and the analysis panels. Eleven automated checks drive the real application headlessly and all pass. That test exists because launching the app proves nothing — the script body doesn't run until a browser connects, and we shipped a bug for exactly that reason.

TRANSITION

"Where does that leave us?"

30 Limitations and Future Work

SCRIPT

> The limitations plainly. No accuracy validation — the largest one. Resolution is uneven: our grid is 300 metres, rainfall is 5.3 kilometres. The distance criteria see only in-district features, affecting eight of our twenty-five sites, though bounded at 13% of the weight. Weights remain expert judgement, and there's no cost dimension. On future work the priority is unambiguous — 150 to 300 structure locations unblock the accuracy figure, the algorithm ranking and a learned model at once.

LIKELY QUESTIONS

Q: Where is the machine learning? A: There's no trained model, and that's deliberate. This is a knowledge-driven model. A learned model needs labelled examples, and no register of structures exists for Chandrapur. Knowledge-driven models work precisely where inventories are unavailable, which is most of rural India — their weakness is subjectivity, and that is what our three sensitivity analyses measure.

TRANSITION

"To close."

31 Conclusion and Next Steps

SCRIPT

Most comparable work produces a map. This produces a map, a ranked shortlist, an explanation for every site, and a measurement of how far the result can be trusted. Established: transferable breakpoints, aggregation reshaping the result far more than weighting does, assigned weight and realised influence differing measurably, and fifteen sites agreed on by all four methods. Not established: no accuracy figure, no cost analysis, no machine-learning result. Being different isn't being better. Thank you.

32 References

SCRIPT

Our references — twelve papers, eight of them GIS-MCDA studies for rainwater harvesting. Happy to take questions.

MOST LIKELY QUESTIONS AFTER THE PRESENTATION

Q1. Which of the four algorithms is best?

Short answer: We can't say without ground truth, and we won't claim it. **Explanation:** Outperformance requires an external standard. We have none, so all we can report is that they differ measurably and where they agree. Fifteen sites were chosen by all four. **Evidence:** Slides 26 and 28; the validation harness returns "missing data".

Q2. How do you know your map is accurate?

Short answer: We verified the software, not the map — and we're explicit about the difference. **Explanation:** Verification asks whether the code does what we specified. Validation asks whether the specification matches reality. We did the first thoroughly and cannot yet do the second. **Evidence:** Eleven automated checks, 99.7% area cross-check, 0.08% river-length conservation, AUC maths verified on five known cases.

Q3. Your weights are subjective. How do you defend them?

Short answer: We don't defend them, we measured them. **Explanation:** A consistency ratio of 0.0235 shows internal coherence, not correctness. So we ran three independent sensitivity analyses. Under ±20% perturbation, 90% of the top-ten sites survive, worst case 70%, maximum rank shift 3. **Evidence:** Slide 27.

Q4. Where is the machine learning? What was your training accuracy?

Short answer: No model is trained. This is a knowledge-driven model, deliberately. **Explanation:** A learned model needs labelled examples. No register of existing structures exists for Chandrapur. Knowledge-driven models are the standard approach precisely where inventories are unavailable. Their weakness is subjectivity, which is what the sensitivity work quantifies. **Evidence:** No ML library is installed; the learned-model comparison is scoped as future work, blocked on the same missing file.

Q5. Where is your train/test split?

Short answer: Nothing learns, so there is nothing to withhold and nothing to overfit. **Explanation:** The parameters come from published thresholds and physical reasoning, not from fitting to data. The equivalent robustness question is "how much does the answer depend on those parameters", which is what we measured. **Evidence:** Slide 27.

Q6. Why these six criteria?

Short answer: Five of them recur across all eight reviewed studies. Settlement proximity we added ourselves. **Explanation:** Rainfall, slope, soil, drainage and land cover are near-universal in this literature. We added settlement proximity because a structure that recharges an aquifer nobody draws from delivers no benefit. **Evidence:** Slides 5, 6 and 13.

Q7. Why is rainfall weighted highest when it barely varies across the district?

Short answer: It's the water source, so it belongs at the top of the hierarchy — and its realised influence genuinely is lower. **Explanation:** Effective-weight analysis measures it at 0.80 times its assigned weight, because percentile scoring fixes its mean near 0.6. We measured that gap rather than assuming it away. **Evidence:** Slide 27, effective weights.

Q8. Why 0.65 and 0.85 for the thresholds?

Short answer: Reasoned, not optimised. **Explanation:** What matters is that they are fixed rather than percentile-based. A percentile split reports about a third of any district as suitable regardless of the data, which makes the number a property of the method. Fixed thresholds make it a measurement. **Evidence:** Slide 16.

Q9. Why a 2.5 km window for drainage density?

Short answer: One kilometre is about three cells and produces speckle; much larger converges to the district mean. **Explanation:** The window has to be wide enough to smooth the effect of whether one channel happens to cross a cell, and narrow enough to still vary across the district. 2.5 km encloses about 20 km². **Evidence:** Slide 22.

Q10. Why 400 people per km² to define a settlement?

Short answer: It classifies 6.6% of the district, which matches the built-up footprint. **Explanation:** It's a reasoned starting value, documented as an assumption rather than an optimised parameter. **Evidence:** Slide 12.

Q11. None of your parameters were optimised. Isn't that a weakness?

Short answer: Correct — none were fitted against an objective function, and we don't claim otherwise. **Explanation:** Each was chosen from physical reasoning and documented. What we did instead was measure how much the result depends on them. That's what the sensitivity analysis reports. **Evidence:** Slide 27; the full parameter table in our methodology documentation.

Q12. Why is your suitable area lower than the published studies?

Short answer: Continuous scoring against fixed breakpoints, and we didn't retune to compensate. **Explanation:** A ramp scores lower than a step class awarding 1.00 across a whole band. Our High share fell from 25.6% to 18.51% when we made that change, and we left the thresholds alone. **Evidence:** Slides 18 and 23.

Q13. All five top sites are Silt Loam cropland. Why?

Short answer: Silt Loam dominates the district and scores 1.00, as does cropland. **Explanation:** It's the model behaving consistently with its tables, but it does concentrate the shortlist in one landscape type — a fair criticism of fully compensatory aggregation. TOPSIS, which penalises weakness harder, still retains nine of our top ten, which suggests it isn't purely an artefact of compensation. **Evidence:** Slides 19 and 26.

Q14. Have you visited any of these sites?

Short answer: No. There is no field verification, and it's listed as a limitation. **Evidence:** Slide 30.

Q15. Isn't gap filling inventing data?

Short answer: It's nearest-neighbour substitution over a few hundred metres, on layers that vary slowly. **Explanation:** It affects at most 2.2% of the district, on the rim only. The alternative was discarding the entire boundary zone, which would have cost more than it saved. **Evidence:** Slide 11.

Q16. What happens if one data layer is missing entirely?

Short answer: Weights renormalise automatically, and we measured the consequence for each layer. **Explanation:** That's exactly what the map-removal analysis does — remove each criterion and re-score. Removing land cover barely changes anything; removing drainage destroys eight of the top ten sites. **Evidence:** Slide 27.

Q17. Why work in latitude and longitude rather than a metric projection?

Short answer: It avoids a second resampling of every layer. **Explanation:** The cost is that cell width changes with latitude, so we compute cell area per raster row. The 99.7% match against the official district area validates that arithmetic. **Evidence:** Slide 10.

Q18. Why 25 sites and 5 km apart?

Short answer: Twenty-five is a shortlist a planner can actually work through; five kilometres ensures they're distinct catchments. **Explanation:** Two structures 500 metres apart harvest the same runoff, so they aren't genuinely different options. **Evidence:** Slides 17 and 19.

Q19. Your drainage density of 0.242 km/km² looks low.

Short answer: HydroRIVERS is thresholded to larger channels, so it understates the network. **Explanation:** Networks derived from a DEM by flow accumulation typically give 0.5 to 2.0. Our class breaks are calibrated to this source, and they would need recalibrating for a DEM-derived network — preserving the class shares rather than the numbers. **Evidence:** Slides 22 and 30.

Q20. You found several bugs in your own system. Doesn't that undermine the results?

Short answer: The opposite — finding them is why the current results can be trusted. **Explanation:** Each was found by checking the implementation against our own documented methodology, and each is now guarded. The ranking defect, for instance, has an automated test. **Evidence:** Slide 20.

Q21. Who provided the expert judgement behind the pairwise matrix?

Short answer: *(Agree this as a team before the review — it is not recorded anywhere in the project.)* **Explanation:** Be prepared to say honestly whether the judgements came from the reviewed literature, from your guide, or from the team's own reasoning.

Q22. What would you do with more time?

Short answer: Ground-truth validation, before anything else. **Explanation:** One dataset of 150 to 300 structure locations unblocks the accuracy figure, the per-algorithm ranking and a learned model simultaneously. The code for all three is already written and waiting. **Evidence:** Slides 28 and 30.

CRITICAL FACTS TO MEMORISE

Everyone should know these regardless of which slides they present. Questions do not respect section boundaries.

Value	What it is
0.0235	Consistency ratio (threshold 0.10)
0.4285 / 0.2254 / 0.1535 / 0.0932 / 0.0597 / 0.0397	Weights: rainfall, slope, soil, drainage, land cover, settlement
18.51% — 2,112.6 km²	High suitability, baseline
54.79% / 26.70%	Medium / Low
123,555	In-district grid cells
450 × 422, ~0.31 × 0.30 km	Analysis grid and cell size
11,412.5 vs 11,443 km² = 99.7%	Area cross-check
0.65 / 0.85	Classification thresholds
80%	Coverage gate
25 sites, 0.984 → 0.866, 5 km apart	The shortlist
1,493 mm/yr, 2014–2024, 11 years	Rainfall
2.5 km	Drainage density window
0.242 km/km²	District drainage density
400 people/km², peak at 3 km	Settlement criterion
90% retained, 70% worst, max shift 3	±20% sensitivity
15 of 36 · 1,417 km² (12.41%)	Consensus across algorithms
18.51 → 13.62 (WLC→TOPSIS) vs 18.51 → 18.32 (AHP→Fuzzy)	The headline finding
11/11	Smoke-test checks passing
97.4% / 97.7%	Slope and population cells above 0.90 under min–max
973 → 1	Cells sharing the maximum score, before and after ramps

ISSUES TO RESOLVE BEFORE PRESENTATION

1. SLIDE 4 CONTRADICTS SLIDE 20 — DECIDE TONIGHT

The inconsistency: Slide 4 presents four LLM/GIS-agent papers and says they provide "the technical foundation for the query-to-action transformation used in this project's spatial analysis component." Slide 20 says the NLP layer was removed from scope. **Which is final:** Slide 20. The component does not exist in the codebase. **Action:** Delete slide 4. It saves 15 seconds you need elsewhere and removes the only self-contradiction in the deck. If you keep it, use the reframing script provided above, and make sure Arya says the same thing on slide 20.

2. CITATION NUMBERING IS INCONSISTENT ACROSS SLIDES 5–7 AND 32

The inconsistency: Slides 5 and 6 label the eight RWH papers [5]–[12], and slide 32's reference list matches that. But slide 7 cites gaps to [1], [2], [3], [4] — under the current numbering those are the LLM papers, which do not discuss AHP weighting or normalisation at all. Slide 7 appears to have been written against a renumbered scheme where the RWH papers are [1]–[8]. **Which is final:** Slide 7's intent is correct; its numbers point at the wrong references. **Action:** If you delete slide 4, renumber the RWH papers to [1]–[8] on slides 5, 6 and 32, and slide 7 becomes correct automatically. If you keep slide 4, fix slide 7's citation numbers instead. A reviewer who checks one citation on slide 7 will find a paper about text-to-SQL.

3. SLIDE 10 — TWO FIGURES ARE NOT FROM THE REPOSITORY

The items: the 27.7 °C mean temperature and the "Census 2011" population figure of 2,204,307. **Action:** Neither is in the project data. Agree a source before presenting, or be ready to say "from the district profile, not from our own data" if asked.

4. TIMING FIGURES WERE NEVER FORMALLY BENCHMARKED

The items: 0.90 s, 0.15 s, 1.5 s, 2.8 s appear on slides 9, 18, 24 and 29. **Action:** These were timed during development, not benchmarked. If asked, say exactly that. Don't present them as measured performance results.

5. WHO PROVIDED THE EXPERT JUDGEMENT

The gap: The provenance of the pairwise comparison matrix is not recorded anywhere in the project. **Action:** Agree the answer as a team before you walk in. This is a very likely question and four different answers would be worse than any single one.

6. SPEAKER SPLIT LEAVES SLIDE 1 UNASSIGNED

Action: Akshay opens, as scripted above. Confirm it so nobody hesitates at the start.